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Smooth Manifold

1.1 Topological Manifolds

Definition 1.1.1

Suppose M is a topological space. We say that M is a **topological manifold of dimension n** or a **topological n -manifold** if it has the following properties:

- M is a **Hausdorff space** (T_2): for every pair of distinct points $p, q \in M$, there are disjoint open subsets $U, V \subseteq M$ such that $p \in U$ and $q \in V$.
- M is **second-countable**: there exists a countable basis for the topology of M .
- M is **locally Euclidean of dimension n** : each point of M has a neighborhood that is homeomorphic to an open subset of $\mathbb{R}^n$.

Remark.

The third property means, more specifically, that for each $p \in M$ we can find

- an open subset $U \subseteq M$ containing p ,
- an open subset $\hat{U} \subseteq \mathbb{R}^n$, and
- a homeomorphism $\varphi : U \rightarrow \hat{U}$.

Remark.

A topological manifold M is locally compact, locally path-connected, separable, Lind  lof, σ -finite, T_3 , T_4 , and paracompact.

1.1.1 Coordinate Chart

Definition 1.1.2 ▶ Coordinate Chart

Let M be a topological n -manifold. A **coordinate chart** (or just a **chart**) on M is a pair (U, φ) , where U is an open subset of M and $\varphi : U \rightarrow \hat{U}$ is a homeomorphism from U to an open subset $\hat{U} = \varphi(U) \subseteq \mathbb{R}^n$.

Remark.

- By definition of a topological manifold, each point $p \in M$ is contained in the domain of some chart (U, φ) .
- If $\varphi(p) = 0$, we say that the chart is **centered at p** .
- If (U, φ) is any chart whose domain contains p , it is easy to obtain a new chart centered at p by subtracting the constant vector $\varphi(p)$.

1.2 Smooth Structures

Definition 1.2.1 ▶ Smoothly Compatible

Let M be a topological n -manifold.

- If $(U, \varphi), (V, \psi)$ are two charts such that $U \cap V \neq \emptyset$, the composite map $\psi \circ \varphi^{-1} : \varphi(U \cap V) \rightarrow \psi(U \cap V)$ is called the **transition map from φ to ψ** .
- Two charts (U, φ) and (V, ψ) are said to be **smoothly compatible** if either $U \cap V = \emptyset$ or the transition map $\psi \circ \varphi^{-1}$ is a diffeomorphism.

Remark.

- $\psi \circ \varphi^{-1}$ is a composition of homeomorphisms, and is therefore itself a homeomorphism.
- Since $\varphi(U \cap V)$ and $\psi(U \cap V)$ are open subsets of $\mathbb{R}^n$, smoothness of this map is to be interpreted in the ordinary sense of having continuous partial derivatives of all orders.

Definition 1.2.2 ▶ Atlas

We define an **atlas for M** to be a collection of charts whose domains cover M . An atlas $\mathcal{A}$ is called a **smooth atlas** if any two charts in $\mathcal{A}$ are smoothly compatible with each other.

Definition 1.2.3 ▶ Maximal Atlas

A smooth atlas $\mathcal{A}$ on M is **maximal** if it is not properly contained in any larger smooth atlas. This just means that any chart that is smoothly compatible with every chart in $\mathcal{A}$ is already in $\mathcal{A}$.

Definition 1.2.4 ▶ Smooth Structures and Smooth Manifolds

- If M is a topological manifold, a **smooth structure on M** is a maximal smooth atlas.
- A **smooth manifold** is a pair $(M, \mathcal{A})$, where M is a topological manifold and $\mathcal{A}$ is a smooth structure on M .

Lemma 1.2.5 ▶ Smooth Manifold Chart Lemma

Let M be a set, and suppose we are given a collection $\{U_\alpha\}$ of subsets of M together with maps $\varphi_\alpha : U_\alpha \rightarrow \mathbb{R}^n$, such that the following properties are satisfied:

1. For each α , φ_α is a bijection between U_α and an open subset $\varphi_\alpha(U_\alpha) \subseteq \mathbb{R}^n$.
2. For each α and β , the sets $\varphi_\alpha(U_\alpha \cap U_\beta)$ and $\varphi_\beta(U_\alpha \cap U_\beta)$ are open in $\mathbb{R}^n$.
3. Whenever $U_\alpha \cap U_\beta \neq \emptyset$, the map $\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \rightarrow \varphi_\beta(U_\alpha \cap U_\beta)$ is smooth.
4. Countably many of the sets U_α cover M .
5. Whenever p, q are distinct points in M , either there exists some U_α containing both p and q or there exist disjoint sets U_α, U_β with $p \in U_\alpha$ and $q \in U_\beta$.

Then M has a unique smooth manifold structure such that each $(U_\alpha, \varphi_\alpha)$ is a smooth chart.

1.3 Manifolds with Boundary

Definition 1.3.1 ▶ Manifolds with Boundary

An **n -dimensional topological manifold with boundary** is a second-countable Hausdorff space M in which every point has a neighborhood homeomorphic either to an open subset of $\mathbb{R}^n$ or to a (relatively) open subset of $\mathbb{H}^n$.

Definition 1.3.2 ► Chart for Manifolds with Boundary

Let M be an n -dimensional topological manifold with boundary. An open subset $U \subseteq M$ together with a map $\varphi : U \rightarrow \mathbb{R}^n$ that is a homeomorphism onto an open subset of $\mathbb{R}^n$ or $\mathbb{H}^n$ will be called a **chart for M** .

- When it is necessary to make the distinction, we will call (U, φ) an **interior chart** if $\varphi(U)$ is an open subset of $\mathbb{R}^n$ (which includes the case of an open subset of $\mathbb{H}^n$ that does not intersect $\partial\mathbb{H}^n$).
- We call (U, φ) a **boundary chart** if $\varphi(U)$ is an open subset of $\mathbb{H}^n$ such that $\varphi(U) \cap \partial\mathbb{H}^n \neq \emptyset$.
- A boundary chart whose image is a set of the form $B_r(x) \cap \mathbb{H}^n$ for some $x \in \partial\mathbb{H}^n$ and $r > 0$ is called a **coordinate half-ball**.